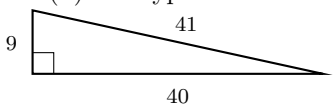
Math 15 — Unit 6: Assignments

The assignment from every lesson in the booklet. Printed work is not tracked on the website.

Lesson 1 Assignment *Show your work.*

Part A — Naming the Sides and Stating the Theorem

1. For the right triangle shown, identify (a) the two legs and (b) the hypotenuse.



(a) Legs: _____
(b) Hypotenuse: _____

2. A right triangle has side lengths 10, 24, and 26. Without drawing a diagram, state which length must be the hypotenuse, and explain how you know.

3. Fill in the blanks: The Pythagorean theorem states that in a right triangle, $___^2 + ___^2 = ___^2$, where c represents the _____.

Part B — Verifying the Theorem and Finding the Hypotenuse

Show both sides of the equation for questions 4 and 5.

4. Verify that a right triangle with legs 18 and 24 and hypotenuse 30 satisfies the Pythagorean theorem.

5. Verify that a right triangle with legs 15 and 20 and hypotenuse 25 satisfies the Pythagorean theorem.

Find the hypotenuse of each right triangle from the two legs given.

6. Legs 20 and 21.

7. Legs 12 and 35.

8. Legs 3 and 6. Give your answer as a simplified mixed radical.

Part C — Finding a Missing Leg

9. Find the missing leg of a right triangle with hypotenuse 61 and one leg 11.

10. Find the missing leg of a right triangle with hypotenuse 34 and one leg 30.

Part D — Applications and Extension

11. A 26 ft ladder leans against a wall with its base 10 ft from the wall. How high up the wall does the ladder reach?

12. A rectangular TV screen is 32 in wide and 24 in tall. A TV's size is the length of the diagonal, corner to corner. Find the size of this TV.

13. *(Multiple Choice)* A right triangle has legs of length 28 and 45. What is the length of the hypotenuse?
A. 53 B. 73 C. 17 D. 1260

14. *(Numerical Response)* A right triangle has a hypotenuse of length 20 and one leg of length 12. The length of the other leg is _____.

15. A triangle has side lengths 6, 7, and 9. Use the Pythagorean theorem to **explain** whether this triangle is a right triangle.

Answer Key. 1. (a) legs 9 and 40 (b) hypotenuse 41. **2.** 26 — it is the longest of the three lengths, and the hypotenuse is the side opposite the right angle. **3.** $a^2 + b^2 = c^2$, where c is the **hypotenuse**. **4.** $18^2 + 24^2 = 324 + 576 = 900$ and $30^2 = 900$; equal, so the theorem holds. **5.** $15^2 + 20^2 = 225 + 400 = 625$ and $25^2 = 625$; equal, so the theorem holds. **6.** 29. **7.** 37. **8.** $3\sqrt{5}$. **9.** 60. **10.** 16. **11.** 24 ft. **12.** 40 in. **13.** A. **14.** 16. **15.** $6^2 + 7^2 = 36 + 49 = 85$, but $9^2 = 81$. Since $85 \neq 81$, it is **not** a right triangle.

Lesson 2 Assignment *Show your work.*

Part A — Finding Lengths From Coordinates

1–3. Determine the length of each line segment.

1(a) $A(2,5)$ to $B(9,5)$

1(b) $C(-3,6)$ to $D(-3,1)$

1(c) $E(4,-2)$ to $F(-5,-2)$

1(d) $G(-1,-7)$ to $H(-1,3)$

2(a) $I(-8,-3)$ to $J(-8,-9)$

2(b) $K(-12,4)$ to $L(5,4)$

2(c) $M(6,-15)$ to $N(6,8)$

2(d) $P(-20,-6)$ to $Q(14,-6)$

3(a) $R(150,-60)$ to $S(150,275)$

3(b) $T(-480,90)$ to $U(365,90)$

3(c) $V(-1200,-50)$ to $W(-1200,730)$

3(d) $X(2075,600)$ to $Y(-925,600)$

4. Determine whether each line segment is horizontal or vertical, and write an expression for its length.

(a) $A(x,y)$ to $B(x+7,y)$

(b) $C(h-3,k+5)$ to $D(h-3,k-9)$

(c) $E(a,b)$ to $F(c,b)$, where $a > c$

(d) $G(s,t)$ to $H(s,w)$, where $w > t$

5. Without doing any calculating, is it possible for the length of a line segment to be negative? Using an example of your own (with your own coordinates), explain why we take the **absolute value** of the difference of the coordinates when computing a length.

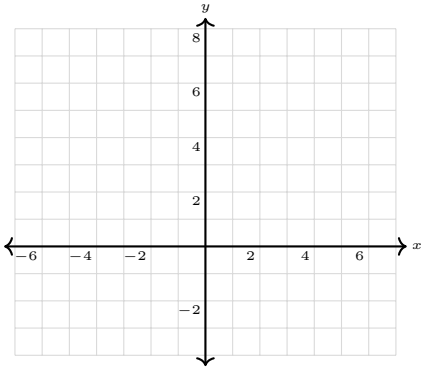
6. Solve for the unknown coordinate.

(a) Segment JK has endpoints $J(3,4)$ and $K(x,4)$. If $JK = 8$ units and $x > 3$, find x.

(b) Segment LM has endpoints $L(-2,y)$ and $M(-2,9)$. If $LM = 13$ units and $y < 9$, find y.

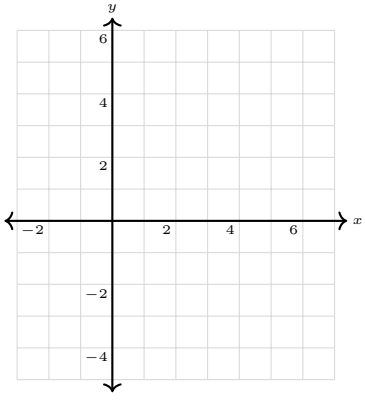
Part B — Grids, Shapes, and Applications

7. Plot the vertices of rectangle $ABCD$: $A(-6,-2)$, $B(5,-2)$, $C(5,6)$, $D(-6,6)$. Determine the perimeter of rectangle $ABCD$.



Perimeter: _____

8. Right triangle PQR has vertices $P(-2,-4)$, $Q(6,-4)$, $R(-2,5)$, with the right angle at P. Plot the triangle and calculate its **area**.



Area: _____

9. A rectangular garden has corners at $(0, 0)$, $(18, 0)$, $(18, 12)$, and $(0, 12)$, measured in metres. Sanjay wants to build a fence all the way around the garden. What total length of fencing does he need?

10. A warehouse wall is being painted. One wall runs from $(0, 0)$ to $(25, 0)$, and the wall it joins runs from $(25, 0)$ to $(25, 14)$, both measured in metres. What is the total length of wall that will be painted?

11. Segment EF has endpoints $E(-10, 3)$ and $F(6, 3)$. Segment GH has endpoints $G(2, -9)$ and $H(2, 19)$. Which segment is longer, and by how many units?

12. *(Multiple Choice)* Horizontal segment CD has endpoints $C(-3, 5)$ and $D(x, 5)$, and its length is 8 units. Which gives all possible values of x?
A. $x = 5$ only B. $x = 11$ only
C. $x = 5$ or $x = -11$ D. $x = 11$ or $x = -11$

13. *(Numerical Response)* Rectangle $WXYZ$ has vertices $W(-6, -4)$, $X(9, -4)$, $Y(9, 7)$, and $Z(-6, 7)$. The perimeter of rectangle $WXYZ$, in units, is _____.

Answer Key. 1. (a) 7 (b) 5 (c) 9 (d) 10. **2.** (a) 6 (b) 17 (c) 23 (d) 34. **3.** (a) 335 (b) 845 (c) 780 (d) 3000. **4.** (a) horizontal, 7 (b) vertical, 14 (c) horizontal, $a - c$ (d) vertical, $w - t$. **5.** No — a length is the *absolute value* of a difference, so it can never be negative; subtracting in the reverse order only flips the sign (examples will vary). **6.** (a) $x = 11$ (b) $y = -4$. **7.** Sides 11 and 8; $P = 2(11) + 2(8) = 38$ units. **8.** Legs $PQ = 8$, $PR = 9$; $A = \frac{1}{2}(8)(9) = 36$ units^2. **9.** $2(18) + 2(12) = 60$ m of fencing. **10.** $25 + 14 = 39$ m of wall. **11.** $EF = 16$, $GH = 28$; GH is longer by 12 units. **12.** C. **13.** 52.

Lesson 3 Assignment *Show your work.*

Part A — Building the Right Triangle

1–3. Determine the run, the rise, and the exact length of each segment. For question 1, build a right triangle under the segment.

1. AB with $A(2, -1)$ and $B(5, 3)$

2. CD with $C(2, 3)$ and $D(10, 9)$

3. EF with $E(-2, 1)$ and $F(1, 5)$

Part B — The Distance Formula

4–7. Use the distance formula to determine the exact length of each segment.

4. $G(0, 0)$ to $H(8, 15)$

5. $I(-4, -3)$ to $J(4, 3)$

6. $K(1, 2)$ to $L(4, 8)$. Give your answer as a mixed radical in simplest form.

7. $M(-1, 3)$ to $N(3, -1)$. Give your answer as a mixed radical in simplest form.

8. Use the distance formula to determine the length of the segment from $P(0, 0)$ to $Q(5, 3)$. Give your answer as a decimal, rounded to the nearest tenth.

9. Explain, in your own words, how the run and the rise used in the distance formula relate to the horizontal and vertical legs of the right triangle formed under a diagonal segment.

Part C — Figures, Maps, and Applications

10. A triangle has vertices $X(1, 1)$, $Y(7, 1)$, and $Z(4, 5)$.
(a) Determine the length of each side of the triangle.
(b) Determine the perimeter of the triangle.

11. A rectangular field has corners at $(0, 0)$ and $(15, 8)$, measured in metres. Determine the exact length of the diagonal path across the field, from one corner to the opposite corner.

12. Use the distance formula to determine the exact length of each segment.
(a) $R(-5, -1)$ to $S(1, 7)$
(b) $T(3, -4)$ to $U(-3, 5)$, rounded to the nearest tenth

13. A town places a coordinate grid over a city map. The school is located at $(2, 3)$ and the library is located at $(10, 9)$, where each grid unit represents 2 km.
(a) Determine the distance between the school and the library, in grid units.
(b) Determine the actual distance between the school and the library, in kilometres.

14. A delivery app places a coordinate grid over a city, with each grid unit representing 1 km. The warehouse is located at $(-5, 3)$ and a customer's house is located at $(2, -5)$. Determine the straight-line distance between the warehouse and the customer's house, to the nearest tenth of a kilometre.

15. *(Multiple Choice)* Which of the following statements about the distance formula $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ is true?

A. It can only be used when the segment is horizontal.
B. It can only be used when the segment is vertical.
C. It can be used for any segment, including diagonal ones, since a horizontal or vertical segment is just a special case where one of the differences is zero.
D. It cannot be used when a coordinate is negative.

Answer Key. 1. run $= 3$, rise $= 4$, $AB = 5$. **2.** run $= 8$, rise $= 6$, $CD = 10$. **3.** run $= 3$, rise $= 4$, $EF = 5$. **4.** $GH = 17$. **5.** $IJ = 10$. **6.** $KL = 3\sqrt{5}$. **7.** $MN = 4\sqrt{2}$. **8.** $PQ \approx 5.8$. **9.** The run, $x_2 - x_1$, is the horizontal leg of the right triangle; the rise, $y_2 - y_1$, is the vertical leg. The distance formula is just $c = \sqrt{a^2 + b^2}$ with the run and rise in place of a and b. **10.** (a) $XY = 6$, $XZ = 5$, $YZ = 5$ (b) perimeter $= 16$ units. **11.** 17 m. **12.** (a) $RS = 10$ (b) $TU = 3\sqrt{13} \approx 10.8$. **13.** (a) 10 grid units (b) 20 km. **14.** ≈ 10.6 km. **15.** C.

Lesson 4 Assignment *Show your work.*

Part A — Finding Midpoints

1. Each segment is horizontal or vertical. Without using the midpoint formula, determine the midpoint by counting squares (imagine each pair of points plotted on a grid).
(a) Horizontal segment: $A(-7, 3)$, $B(1, 3)$. (b) Vertical segment: $C(4, 7)$, $D(4, -5)$.

2. Use the midpoint formula to determine the midpoint of the segment with each pair of endpoints.
(a) $A(3, 5)$, $B(9, 11)$ (b) $C(2, 10)$, $D(8, 4)$ (c) $E(1, 7)$, $F(11, 3)$ (d) $G(6, 2)$, $H(0, 14)$

3. Determine the midpoint of the segment with each pair of endpoints. Each pair involves at least one negative coordinate.
(a) $A(-4, 6)$, $B(2, -8)$ (b) $C(-10, -3)$, $D(6, -11)$ (c) $E(-7, 9)$, $F(-1, -5)$ (d) $G(5, -6)$, $H(-11, -12)$

4. Determine the midpoint of the segment with each pair of endpoints. 5. Mixed review: determine the midpoint of the segment with each pair of endpoints.
4(a) $P(-2.5, 3.6)$, $Q(4.5, -7.6)$ 4(b) $R(5.4, -2.2)$, $S(-1.8, 6.6)$ 5(a) $A(8, 1)$, $B(2, 9)$ 5(b) $C(-9, 4)$, $D(-1, -10)$ 5(c) $E(3.2, -4.8)$, $F(-1.2, 0.8)$

Part B — Explaining, Checking and Applying

6. In your own words, explain what the midpoint of a line segment is. Then explain why averaging the two endpoints' x-coordinates gives the x-coordinate of the midpoint.

7. A city planner claims that the midpoint of the segment joining $S(-6, 10)$ and $T(8, -2)$ is $(2, 4)$. Use the midpoint formula to check the claim. If it is incorrect, state the correct midpoint.

8. Two trailheads, Riverside at $(-18, 12)$ and Summit View at $(24, -6)$, are joined by a straight trail on a map where each unit is 1 km. Rangers want a rest shelter exactly halfway between them. Determine its coordinates.

9. Jasmine's house is 6 blocks east and 10 blocks north of the community centre. Tyler's house is 8 blocks west and 2 blocks south of the community centre. Using the community centre as the origin, determine the coordinates of the point exactly halfway between their houses.

10. *(Multiple Choice)* The midpoint of the segment joining $(2, -3)$ and $(10, 7)$ is
A. $(4, 5)$ B. $(6, 2)$ C. $(12, 4)$ D. $(8, 10)$

11. *(Multiple Choice)* Which pair of points has a midpoint at the origin, $(0, 0)$?
A. $(3, 5)$ and $(3, -5)$ B. $(3, 5)$ and $(-3, 5)$
C. $(3, 5)$ and $(-3, -5)$ D. $(3, 5)$ and $(5, 3)$

12. *(Multiple Choice)* Which of the following correctly calculates the midpoint of a segment with endpoints (x_1, y_1) and

(x_2, y_2)? A. $(x_2 - x_1,\ y_2 - y_1)$ B. $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ C. $\left(\frac{x_1+y_1}{2},\ \frac{x_2+y_2}{2}\right)$ D. $\left(\frac{x_1+x_2}{2},\ \frac{y_1+y_2}{2}\right)$

13. *(Numerical Response)* The midpoint of the segment joining $(-9, 4)$ and $(15, -12)$ has x-coordinate _____.

14. *(Numerical Response)* The midpoint of the segment joining $(6, 17)$ and $(-2, -5)$ has y-coordinate _____.

15. *(Numerical Response)* $E(-4, 9)$ and $F(10, -3)$ are two corners of a garden, each unit 1 m. A sprinkler sits at the midpoint of $\overline{EF}$. The sum of its x- and y-coordinates is _____.

Answer Key. 1. (a) $(-3, 3)$ (b) $(4, 1)$. **2.** (a) $(6, 8)$ (b) $(5, 7)$ (c) $(6, 5)$ (d) $(3, 8)$. **3.** (a) $(-1, -1)$ (b) $(-2, -7)$ (c) $(-4, 2)$ (d) $(-3, -9)$. **4.** (a) $(1, -2)$ (b) $(1.8, 2.2)$. **5.** (a) $(5, 5)$ (b) $(-5, -3)$ (c) $(1, -2)$. **6.** The point exactly halfway between the endpoints, the same distance from each; averaging works because the average of two numbers lands exactly halfway between them, so the midpoint's x-position is halfway between the endpoints' x-positions. **7.** Incorrect — the midpoint is $(1, 4)$. **8.** $(3, 3)$. **9.** $(-1, 4)$. **10.** B **11.** C **12.** D **13.** 3 **14.** 6 **15.** 6.

Lesson 5 Assignment *Show your work.*

Part A — Finding a Missing Endpoint

1. In each case, one endpoint and the midpoint of a segment are given. Determine the coordinates of the other endpoint.

(a) $P(3, 2)$ is an endpoint of $\overline{PQ}$, with midpoint $M(8, 6)$. Find Q.

(b) $A(6, 9)$ is an endpoint of $\overline{AB}$, with midpoint $M(10, 4)$. Find B.

2. Same idea, but the coordinates involve negative numbers — the method is the same, just be careful adding signed numbers.

(a) $C(-5, -3)$ is an endpoint of $\overline{CD}$, with midpoint $M(0, 2)$. Find D.

(b) $E(-8, 6)$ is an endpoint of $\overline{EF}$, with midpoint $M(-2, -1)$. Find F.

3. A canal has a lock built at its midpoint, $M(5, 9)$. One end of the canal, at the reservoir, is located at $R(1, 3)$. Determine the coordinates of the harbour, H, at the other end of the canal.

4. An underground pipeline runs from a pumping station to a storage tank. A shut-off valve sits at the pipeline's midpoint, $M(-4, 7)$. The pumping station is at $P(-10, 1)$. Determine the coordinates of the storage tank, T.

Part B — Distance and Map Applications

5. Determine the exact distance between each pair of points.

(a) $A(1, 2)$ and $B(9, 8)$

(b) $C(-3, 5)$ and $D(5, -10)$

6. Determine the distance between $E(2, 3)$ and $F(9, -4)$, to the nearest tenth.

7. A ferry travels from a dock at $(2, 5)$ to an island at $(14, 10)$ on a lake map, where each grid unit represents 300 m. Determine the length of the ferry's route, in metres.

8. An airplane's flight path is tracked from an airport at $(-10, -5)$ to a beacon tower at $(14, 2)$, where each grid unit represents 3 km. Determine the flight distance, in kilometres.

Part C — Midpoint Applications

9. Two friends, Jasmine at $J(-8, 5)$ and Kyle at $K(10, -3)$, want to meet at a coffee shop exactly halfway between their houses (1 grid unit represents 1 block). Determine the coordinates of the coffee shop.

10. A straight road connects Elm Junction at $(-14, 20)$ and Birch Junction at $(22, -8)$. A rest stop is to be built at the midpoint of the road. Determine the coordinates of the rest stop.

Part D — Combining Both Formulas

11. A hiking trail begins at trailhead $A(3, 2)$. The midpoint rest bench is located at $M(9, 10)$. Each grid unit represents 25 m.
(a) Determine the coordinates of the far end of the trail, B.
(b) Determine the total length of the trail, in metres.

12. A fence is to be built between two posts at $P(-7, -9)$ and $Q(9, 3)$, where each grid unit represents 5 m.
(a) Determine the coordinates of the midpoint post, M.
(b) Determine the total length of the fence, in metres.

Part E — Multiple Choice and Numerical Response

13. *(Multiple Choice)* The midpoint of $\overline{AB}$ is $M(4, -1)$. If A is $(10, 3)$, the coordinates of B are
A. $(-2, -5)$ B. $(7, 1)$ C. $(-2, 7)$ D. $(2, -5)$

14. *(Multiple Choice)* The distance between the points $(1, 1)$ and $(5, 4)$ is
A. 5 B. 7 C. $\sqrt{7}$ D. 25

15. *(Numerical Response)* $M(6, 2)$ is the midpoint of $\overline{CD}$. If C is $(-2, -4)$, the y-coordinate of D is _____.

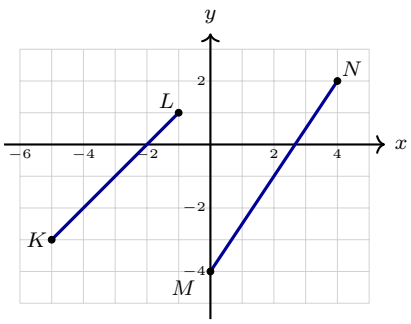
16. *(Numerical Response)* The distance between $(0, 0)$ and $(9, 12)$ is _____ units.

Answer Key. 1. (a) $Q(13, 10)$ (b) $B(14, -1)$. **2.** (a) $D(5, 7)$ (b) $F(4, -8)$. **3.** $H(9, 15)$. **4.** $T(2, 13)$. **5.** (a) 10 (b) 17. **6.** 9.9. **7.** 3900 m. **8.** 75 km. **9.** $(1, 1)$. **10.** $(4, 6)$. **11.** (a) $B(15, 18)$ (b) 500 m. **12.** (a) $M(1, -3)$ (b) 100 m. **13.** A. **14.** A. **15.** 8. **16.** 15.

Lesson 6 Assignment *Show your work.*

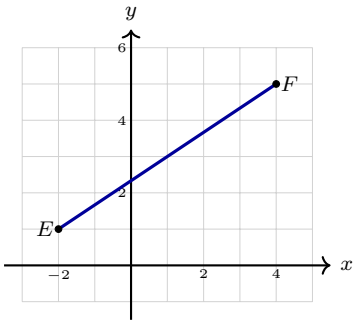
Part A — Counting Rise and Run from a Grid

1. Complete the table by counting rise and run for each segment, then writing its slope.



Segment	Rise	Run	Slope
KL			
MN			

2. Segment EF is plotted below, with $E(-2, 1)$ and $F(4, 5)$.



(a) What is the rise from E to F?

(b) What is the run from E to F?

(c) What is the slope of $\overline{EF}$? Simplify if possible.

Part B — Using the Slope Formula

3. Use the slope formula to determine the slope of the segment with each pair of endpoints.

(a) $P(2, 3),\ Q(5, 9)$ (b) $R(1, 2),\ S(4, 11)$ (c) $T(2, 5),\ U(10, 9)$ (d) $V(1, 1),\ W(9, 7)$

4. Same instruction, but each pair involves negative coordinates. Simplify each answer.

(a) $A(-5, -3),\ B(1, 3)$ (b) $C(-4, -6),\ D(4, -2)$ (c) $E(-2, -7),\ F(6, -1)$ (d) $G(-8, -4),\ H(-2, 5)$

Part C — Applications and Explanations

5. A wheelchair ramp climbs from the ground at $(0, 0)$ to a doorway platform at $(18, 3)$, where each unit represents 1 foot. Determine the slope of the ramp.

6. A road sign warns of a hill ahead. The hill rises 50 m in elevation over a horizontal distance of 250 m. Determine the slope (grade) of the hill as a fraction in lowest terms, and also as a percent.

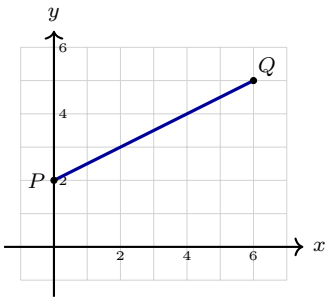
7. In your own words, explain what slope measures, and explain why we call it "rise over run."

8. A student claims the slope of the segment joining $(2, 3)$ and $(10, 9)$ is $\frac{4}{3}$. Use the slope formula to check the claim. If it is incorrect, state the correct slope and explain the mistake the student most likely made.

9. A staircase has steps that are all the same size. Each step has a rise of 20 cm and a run of 25 cm. Determine the slope of the line connecting the bottom of the first step to the top of the last step.

10. Two survey stakes are placed on a hillside at $(10, 20)$ and $(570, 160)$, where each unit represents 1 metre. Determine the slope of the hillside between the two stakes, in lowest terms.

Part D — Multiple Choice and Numerical Response



11. *(Multiple Choice)* The slope of $\overline{PQ}$ shown at left is

A. $\frac{1}{2}$ B. 2 C. 3 D. 6

12. *(Multiple Choice)* Which point, together with $(2, 4)$, forms a line segment with a slope of 3?

A. $(3, 7)$ B. $(3, 1)$ C. $(5, 4)$ D. $(5, 5)$

13. *(Multiple Choice)* For points (x_1, y_1) and (x_2, y_2), the slope formula is

A. $m = \dfrac{x_2 - x_1}{y_2 - y_1}$ B. $m = \dfrac{y_2 - y_1}{x_2 - x_1}$ C. $m = \dfrac{y_2 + y_1}{x_2 + x_1}$

D. $m = \dfrac{y_1 - y_2}{x_2 - x_1}$

14. *(Numerical Response)* The slope of the segment joining $(3, 5)$ and $(11, 9)$ is _____.

15. *(Numerical Response)* The slope of the line through $(-6, -9)$ and $(2, -1)$ is _____.

Answer Key. 1. KL: rise 4, run 4, slope 1; MN: rise 6, run 4, slope $\frac{3}{2}$. **2.** (a) 4 (b) 6 (c) $\frac{2}{3}$. **3.** (a) 2 (b) 3 (c) $\frac{1}{2}$ (d) $\frac{3}{4}$. **4.** (a) 1 (b) $\frac{1}{2}$ (c) $\frac{3}{4}$ (d) $\frac{3}{2}$. **5.** $\frac{1}{6}$. **6.** $\frac{1}{5}$, or 20%. **7.** Slope measures steepness: it compares vertical change (rise) to horizontal change (run), so "rise over run" names the ratio being calculated. **8.** Incorrect; correct slope is $\frac{3}{4}$ (the student swapped rise and run, putting the run on top). **9.** $\frac{4}{5}$. **10.** $\frac{1}{4}$. **11.** A ($m = \frac{3}{6} = \frac{1}{2}$). **12.** A. **13.** B. **14.** 0.5. **15.** 1.

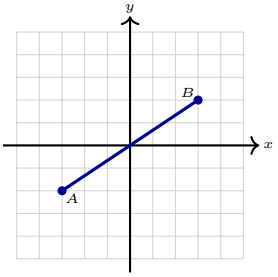
Lesson 7 Assignment *Show your work.*

Part A — Vocabulary and One Type at a Time

1. Fill in each blank.
(a) A line that rises from left to right has a _____ slope.
(b) A line that falls from left to right has a _____ slope.
(c) A perfectly horizontal line has a slope of _____.
(d) A perfectly vertical line has a(n) _____ slope.

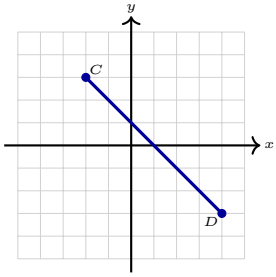
2–5. Calculate the slope of each segment below. For questions 2 and 3, also state whether the slope is positive or negative.

2. AB: $A(-3,-2)$, $B(3,2)$



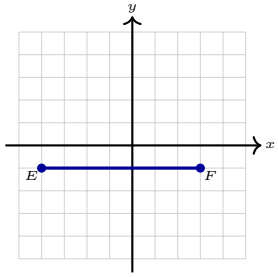
Slope: _____
Type: _____

3. CD: $C(-2,3)$, $D(4,-3)$



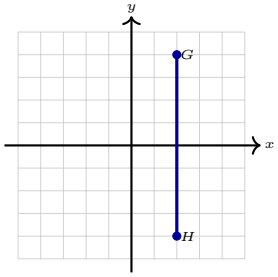
Slope: _____
Type: _____

4. EF: $E(-4,-1)$, $F(3,-1)$



Slope: _____

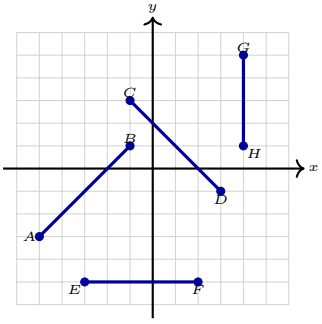
5. GH: $G(2,4)$, $H(2,-4)$



Slope: _____

Part B — Mixing the Four Types

6. Each segment has integer endpoints: $A(-5,-3)$ and $B(-1,1)$; $C(-1,3)$ and $D(3,-1)$; $E(-3,-5)$ and $F(2,-5)$; $G(4,5)$ and $H(4,1)$. Complete the table.



Segment	Rise	Run	Slope	Type
AB				
CD				
EF				
GH				

7. State whether each equation describes a horizontal or a vertical line, and give the slope of that line.

(a) $y = 5$
Line: _____
Slope: _____

(b) $x = -2$
Line: _____
Slope: _____

(c) $y = -7$
Line: _____
Slope: _____

(d) $x = 0$
Line: _____
Slope: _____

8. In your own words, explain the difference between a slope of 0 and an undefined slope. Use the words *horizontal*, *vertical*, and *divide* somewhere in your answer.

9. Calculate the slope of the line through each pair of points, and state its type (positive, negative, zero, or undefined).

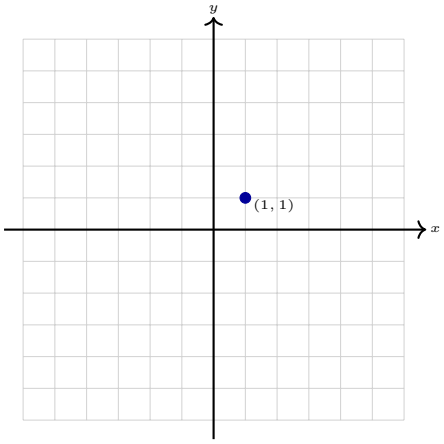
(a) $(1,2)$ and $(4,8)$
(b) $(5,-1)$ and $(2,5)$
(c) $(-3,6)$ and $(4,6)$
(d) $(7,-2)$ and $(7,9)$

Part C — Applications, Explanations, and Practice

10. A ski hill is measured against a grid laid over a side-view photo. A skier starts at $(0, 10)$ at the top of the run and stops at $(40, 0)$ at the bottom, where each unit is 1 m. Calculate the slope of the hill along this line, and explain in words why it makes sense for this slope to be negative.

11. Explain, using the words *rise*, *run*, and *divide*, why the slope of a vertical line is undefined rather than equal to 0.

12. On the grid, draw four lines through $(1, 1)$: one with positive slope, one with negative, one with zero, and one with undefined slope. Label each line with its type.



13. *(Multiple Choice)* Which of the following best describes the slope of a flat, level road?
A. positive B. negative C. zero D. undefined

14. *(Multiple Choice)* A line segment has endpoints $(3, -2)$ and $(3, 5)$. Its slope is
A. 0 B. 7 C. undefined D. -7

15. *(Numerical Response)* The points $(-2, 4)$ and $(6, 4)$ lie on a line. The slope of this line is _____.

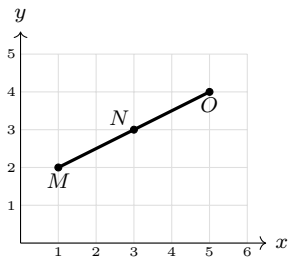
Answer Key. 1. (a) positive (b) negative (c) 0 (d) undefined. **2.** $\frac{4}{6} = \frac{2}{3}$, positive. **3.** $\frac{-6}{6} = -1$, negative. **4.** 0. **5.** undefined. **6.** AB: rise 4, run 4, slope 1, positive; CD: rise -4, run 4, slope -1, negative; EF: rise 0, run 5, slope 0, zero; GH: rise -4, run 0, slope undefined, undefined. **7.** (a) horizontal, 0 (b) vertical, undefined (c) horizontal, 0 (d) vertical, undefined. **8.** A slope of 0 belongs to a *horizontal* line: the rise is 0, and 0 divided by any run is 0. An undefined slope belongs to a *vertical* line: the run is 0, and you cannot *divide* by 0, so no number exists. **9.** (a) $\frac{6}{3} = 2$, positive (b) $\frac{6}{-3} = -2$, negative (c) $\frac{0}{7} = 0$, zero (d) $\frac{11}{0}$, undefined. **10.** $\frac{0-10}{40-0} = -\frac{1}{4}$; the elevation ($y$) drops as the horizontal distance from the top (x) increases, so the line falls left to right. **11.** Every point on a vertical line has the same x, so the *run* between any two points is 0; slope *divides* the *rise* by the run, and dividing by 0 has no answer — so the slope is undefined, not 0. **12.** Answers vary; check that the positive line rises, the negative line falls, the zero-slope line is horizontal and the undefined-slope line is vertical, all through $(1, 1)$. **13.** C. **14.** C. **15.** 0.

Lesson 8 Assignment *Show your work.*

Part A — Reading and Testing for Collinearity

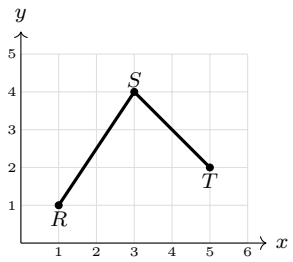
1. Each grid shows three labelled points joined by straight segments. By looking only at the picture, state whether the three points appear to be collinear.

(a)



Answer: _____

(b)



Answer: _____

2. Use the slope test to determine whether $A(0,1)$, $B(2,5)$, and $C(4,9)$ are collinear.

3. Use the slope test to determine whether each set of points is collinear.
(a) $P(1,3)$, $Q(4,9)$, $R(7,15)$ (b) $S(-2,0)$, $T(0,4)$, $U(3,9)$ (c) $K(2,-3)$, $L(5,3)$, $M(8,9)$

Part B — Negative Coordinates and Points on a Segment

4. The points $D(-6,-5)$, $E(-2,-1)$, and $F(2,3)$ all involve negative coordinates. Determine whether D, E, and F are collinear.

5. The points $G(-4,2)$, $H(-1,-1)$, and $I(3,-4)$ all involve negative coordinates. Determine whether G, H, and I are collinear.

6. Consider the points $A(1,2)$ and $C(9,14)$.
(a) Does $D(5,8)$ lie on line segment AC? Use slope to justify your answer.

(b) Does $E(5,9)$ lie on line segment AC? Use slope to justify your answer.

Part C — Unknown Coordinates and Applications

7. The points $J(3,5)$, $K(6,11)$, and $L(9,k)$ are collinear. Find the value of k.

8. The points $P(1,2)$, $Q(5,y)$, and $R(7,11)$ are collinear. Find the value of y.

9. A surveyor marks three fence posts at $A(0,0)$, $B(6,4)$, and $C(15,10)$ on a site map, each unit 1 m. Do the three posts lie along the same straight fence line? Justify using slope.

10. A courier's first three stops are $D_1(1,3)$, $D_2(5,9)$, and $D_3(9,14)$ on a city map, each unit 1 km. Can the courier drive one straight line through all three? Justify using slope.

11. In your own words, explain why calculating m_{AB} and m_{BC} (using the shared point B) is enough to prove A, B, and C are collinear. What would $m_{AB} \neq m_{BC}$ mean?

12. *(Mixed Review)* Consider $A(-8,-3)$, $B(-2,1)$, $C(4,5)$, and $D(10,9)$.
(a) Show that A, B, and C are collinear.
(b) Does D also lie on this same line? Justify your answer using slope.

Part D — Multiple Choice and Numerical Response

13. *(MC)* The points $E(2,1)$, $F(5,7)$, and $G(8,k)$ are collinear. The value of k is
A. 10 B. 11 C. 13 D. 16

14. *(MC)* Which set of points below is collinear?
A. $(0,0)$, $(2,3)$, $(4,7)$ B. $(1,1)$, $(3,5)$, $(5,9)$
C. $(2,2)$, $(4,5)$, $(6,9)$ D. $(0,3)$, $(2,4)$, $(5,5)$

15. *(NR)* The points $H(-3,-2)$, $I(1,2)$, and $J(6,k)$ are collinear. The value of k is _____.

16. *(NR)* To the nearest tenth, the slope of the line segment joining $C(-4,-9)$ and $D(11,22)$ is _____.

Answer Key. 1. (a) appear collinear (b) do not appear collinear. **2.** $m_{AB} = m_{BC} = 2$, collinear. **3.** (a) $m_{PQ} = m_{QR} = 2$, collinear (b) $m_{ST} = 2$, $m_{TU} = \frac{5}{3}$, not collinear (c) $m_{KL} = m_{LM} = 2$, collinear. **4.** $m_{DE} = m_{EF} = 1$, collinear. **5.** $m_{GH} = -1$, $m_{HI} = -\frac{3}{4}$, **not** collinear. **6.** (a) yes, $m_{AD} = m_{AC} = \frac{3}{2}$ (b) no, $m_{AE} = \frac{7}{4} \neq \frac{3}{2}$. **7.** $k = 17$. **8.** $y = 8$. **9.** yes, $m_{AB} = m_{BC} = \frac{2}{3}$, same fence line. **10.** no, $m_{D_1D_2} = \frac{3}{2} \neq \frac{5}{4} = m_{D_2D_3}$. **11.** Equal slopes means AB and BC point the same way; sharing B, they form one straight line. Unequal slopes means the path bends at B — not collinear. **12.** (a) $m_{AB} = m_{BC} = \frac{2}{3}$, collinear (b) yes, $m_{CD} = \frac{2}{3}$ too. **13.** C. **14.** B. **15.** $k = 7$. **16.** 2.1 (exactly $\frac{31}{15} \approx 2.07$).

Lesson 9 Assignment *Show your work.*

Part A — Slope Review and Parallel Segments

1. Use the slope formula to calculate the slope of the segment with the given endpoints.
(a) $A(2,5)$, $B(6,13)$ (b) $C(-3,4)$, $D(5,-4)$ (c) $E(-6,-2)$, $F(-6,7)$ (d) $G(-8,3)$, $H(9,3)$

2. For each pair of segments, calculate both slopes and state whether the segments are parallel.
(a) $J(1,1)$, $K(4,7)$ and $L(-2,-3)$, $M(1,3)$ (b) $N(-5,6)$, $P(3,2)$ and $Q(-1,5)$, $R(3,-3)$ (c) $S(-4,-5)$, $T(2,7)$ and $U(0,-6)$, $V(3,0)$

Part B — Perpendicular Segments and Negative Reciprocals

3. For each pair of segments, calculate both slopes and state whether the segments are perpendicular.
(a) $W(-3,1)$, $X(1,4)$ and $Y(2,5)$, $Z(5,1)$ (b) $A_1(-2,-3)$, $B_1(4,1)$ and $C_1(0,2)$, $D_1(3,-2)$ (c) $E_1(-5,0)$, $F_1(0,3)$ and $G_1(1,-4)$, $H_1(4,-9)$

4. For each slope, use "flip it, and change the sign" to state the slope of a *perpendicular* line. Also state the slope of a *parallel* line.
(a) $\frac{2}{3}$ (b) -5 (c) $\frac{4}{9}$ (d) $-\frac{3}{7}$

5. Determine whether each pair of slopes describes parallel lines, perpendicular lines, or neither.
(a) $m_1 = \frac{5}{8}$, $m_2 = \frac{5}{8}$ (b) $m_1 = -\frac{2}{3}$, $m_2 = \frac{3}{2}$ (c) $m_1 = 4$, $m_2 = -\frac{1}{4}$

(d) $m_1 = \frac{7}{10}$, $m_2 = \frac{10}{7}$ (e) $m_1 = -6$, $m_2 = -6$ (f) $m_1 = \frac{3}{5}$, $m_2 = -\frac{3}{5}$

Part C — Solving for an Unknown Slope

6. The slopes of two **parallel** lines are given. Determine the value of the variable.
(a) 4, $\frac{k}{3}$ (b) -2, $\frac{n}{5}$ (c) $\frac{3}{4}$, $\frac{p}{8}$ (d) $-\frac{5}{6}$, $-\frac{w}{12}$

7. The slopes of two **perpendicular** lines are given. Determine the value of the variable.
(a) $\frac{1}{2}$, $2h$ (b) 4, $\frac{8}{p}$ (c) -5, $\frac{s}{5}$ (d) $-\frac{3}{4}$, $-\frac{w}{12}$

Part D — Applications and Mixed Practice

8. A resort's trail map shows two ski trails. Trail 1 runs from $(2, 4)$ to $(10, 16)$. Trail 2 runs from $(-3, -2)$ to $(5, 10)$, where each grid unit represents 100 m. Are the two trails parallel?

9. *(Multiple Choice)* A segment passes through $(1, 2)$ and $(5, 10)$. A line perpendicular to this segment has slope
A. 2 B. -2 C. $\frac{1}{2}$ D. $-\frac{1}{2}$

10. *(Multiple Choice)* A segment passes through $(-4, -1)$ and $(2, 5)$. A line parallel to this segment has slope
A. 1 B. -1 C. 6 D. 0

11. Triangle XYZ has vertices $X(-2, -1)$, $Y(2, 3)$, and $Z(6, -1)$. Use slopes to determine whether the triangle is right-angled at Y.

12. Segment UV has endpoints $U(-2, 3)$ and $V(6, k)$. Segment PQ has endpoints $P(1, 5)$ and $Q(5, 3)$. If UV is perpendicular to PQ, find the value of k.

13. Two zip-lines are set up in a park. Zip-line A runs from $(0, 10)$ to $(8, 2)$. Zip-line B runs from $(2, 1)$ to $(6, 5)$, where each unit represents 1 m. Are the two zip-lines perpendicular?

14. For each pair of segments, calculate both slopes and state whether the segments are parallel, perpendicular, or neither.
(a) $A(-6, 2)$, $B(-2, 5)$ and $C(1, -3)$, $D(5, 0)$
(b) $E(-4, -3)$, $F(0, 5)$ and $G(2, 4)$, $H(6, 2)$
(c) $I(-5, -1)$, $J(3, 3)$ and $K(-2, 4)$, $L(2, 0)$

15. *(Multiple Choice)* Which pair of slopes represents **perpendicular** lines? A. $\frac{3}{5}$, $\frac{3}{5}$ B. $\frac{2}{7}$, $\frac{7}{2}$ C. -4, $\frac{1}{4}$ D. $\frac{5}{9}$, $\frac{9}{5}$

Answer Key. 1. (a) 2 (b) -1 (c) undefined (d) 0. **2.** (a) $m_{JK} = 2 = m_{LM}$, parallel (b) $-\frac{1}{2}$ and -2, not parallel (c) $m_{ST} = 2 = m_{UV}$, parallel. **3.** (a) $\frac{3}{4}$, $-\frac{4}{3}$, product -1, perpendicular (b) $\frac{2}{3}$, $-\frac{4}{3}$, product $-\frac{8}{9}$, not perpendicular (c) $\frac{3}{5}$, $-\frac{5}{3}$, product -1, perpendicular. **4.** perp. / parallel: (a) $-\frac{3}{2}$ / $\frac{2}{3}$ (b) $\frac{1}{5}$ / -5 (c) $-\frac{9}{4}$ / $\frac{4}{9}$ (d) $\frac{7}{3}$ / $-\frac{3}{7}$. **5.** (a) parallel (b) perpendicular (c) perpendicular (d) neither (e) parallel (f) neither. **6.** (a) $k = 12$ (b) $n = -10$ (c) $p = 6$ (d) $w = 10$. **7.** (a) $h = -1$ (b) $p = -32$ (c) $s = 1$ (d) $w = -16$. **8.** $m_1 = \frac{3}{2} = m_2$, parallel. **9.** D. **10.** A. **11.** $m_{XY} = 1$, $m_{YZ} = -1$, product -1, so yes — right-angled at Y. **12.** $k = 19$. **13.** $m_A = -1$, $m_B = 1$, product -1, perpendicular. **14.** (a) both $\frac{3}{4}$, parallel (b) 2 and $-\frac{1}{2}$, product -1, perpendicular (c) $\frac{1}{2}$ and -1, neither. **15.** C.

Lesson 10 Assignment *Show your work.*

Part A — Pythagorean Theorem and Segment Lengths

1. Determine the length of the hypotenuse of a right triangle with legs 18 and 24.

2. Determine the length of the missing leg of a right triangle with hypotenuse 37 and one leg 35.

3. A guy-wire anchors a 20 m radio tower to the ground at a point 9 m from the base. Determine the length of the guy-wire, to the nearest tenth of a metre.

4. Determine the length of the horizontal segment joining $E(-9,-2)$ and $F(6,-2)$.

5. Determine the length of the vertical segment joining $G(-4,11)$ and $H(-4,-7)$.

Part B — The Distance Formula

6. Determine the exact distance between $A(2,1)$ and $B(7,13)$.

7. Determine the exact distance, as a simplified mixed radical, between $C(-6,2)$ and $D(0,-4)$.

8. A delivery drone flies from a warehouse at $(-3,2)$ to a depot at $(9,11)$ on a city grid, where each grid unit represents 100 m. Determine the real-world distance from the warehouse to the depot.

Part C — Midpoint, Missing Endpoints, and Slope

9. Determine the coordinates of the midpoint of the segment joining $J(-7,4)$ and $K(9,-10)$.

10. Determine the coordinates of the midpoint of the segment joining $L(3.4,-2.6)$ and $N(8.6,7.4)$.

11. $M(-2,6)$ is the midpoint of segment $\overline{PQ}$. If $P(-9,1)$, determine the coordinates of Q.

12. Two community centres are located at $(-4,10)$ and $(22,-6)$ on a town-planning grid, where each grid unit represents 20 m. A new playground will be built exactly halfway between them. Determine the coordinates of the playground, and state how far east and how far south it is from the first community centre.

13. Determine the slope of the segment joining $P(-5,-4)$ and $Q(7,2)$.

14. Determine the slope of the segment joining $R(4,9)$ and $S(-2,-3)$.

15. A wheelchair ramp rises 1.2 m over a horizontal run of 9.6 m. Determine the slope of the ramp as a decimal.

Part D — Types of Slope, Collinearity, Parallel and Perpendicular

Classify the slope of each segment in questions 16–18 as positive, negative, zero, or undefined.

16. $A(-6, 3)$ and $B(2, 9)$

17. $C(-4, -5)$ and $D(6, -5)$

18. $E(8, 3)$ and $F(8, -7)$

19. Determine whether the points $G(-6, -7)$, $H(-1, -1)$, and $I(9, 11)$ are collinear.

20. The points $J(-3, 2)$, $K(1, 5)$, and $L(9, q)$ are collinear. Determine the value of q.

21. Segment AB joins $A(-5, -3)$ and $B(3, 5)$. Segment CD joins $C(-2, 6)$ and $D(6, 14)$. Are AB and CD parallel, perpendicular, or neither?

22. Segment EF joins $E(-4, 5)$ and $F(2, -3)$. Segment GH joins $G(-1, -2)$ and $H(7, 4)$. Are EF and GH perpendicular?

23. Determine the slope of a line that is perpendicular to a segment with slope $-\dfrac{2}{7}$.

Part E — Mixed Review

24. Triangle ABC has vertices $A(-3, -5)$, $B(3, 3)$, and $C(-5, 9)$.

(a) Determine the slope of AB and the slope of BC, and show that angle B is a right angle.

(b) Determine the length of each side of the triangle using the distance formula.

(c) Determine the midpoint of the hypotenuse, AC.

25. Quadrilateral $EFGH$ has vertices $E(-7, -3)$, $F(-3, 5)$, $G(5, 1)$, and $H(1, -7)$. Determine the slope and length of each side, and identify the most precise name for quadrilateral $EFGH$.

26. Segment JK joins $J(-6, -2)$ and $K(4, q)$. Segment LM joins $L(-5, 7)$ and $M(3, -5)$. Determine the value of q so that $JK \parallel LM$, and then determine the exact length of JK.

Answer Key. **1.** 30. **2.** 12. **3.** ≈ 21.9 m. **4.** 15. **5.** 18. **6.** 13. **7.** $6\sqrt{2}$. **8.** 1500 m. **9.** $(1, -3)$. **10.** $(6, 2.4)$. **11.** $Q(5, 11)$. **12.** $(9, 2)$; 13 units (260 m) east and 8 units (160 m) south. **13.** $\frac{1}{2}$. **14.** 2. **15.** 0.125. **16.** Positive. **17.** Zero. **18.** Undefined. **19.** Collinear ($m_{GH} = m_{HI} = \frac{6}{5}$). **20.** $q = 11$. **21.** Parallel (both $m = 1$). **22.** Perpendicular ($m_{EF} m_{GH} = -\frac{4}{3} \cdot \frac{3}{4} = -1$). **23.** $\frac{7}{2}$. **24.** (a) $m_{AB} = \frac{4}{3}$, $m_{BC} = -\frac{3}{4}$, product -1 so $\angle B = 90°$; (b) $AB = 10$, $BC = 10$, $AC = 10\sqrt{2}$; (c) $(-4, 2)$. **25.** Slopes $2, -\frac{1}{2}, 2, -\frac{1}{2}$ (opposite sides parallel, adjacent perpendicular); all four lengths $4\sqrt{5}$; a **square**. **26.** $q = -17$; $JK = 5\sqrt{13}$.